

Literature Benchmark Brief: Zhou_2022_SciData

Title: SDWPF: A large-scale spatio-temporal wind power forecasting benchmark with dynamic status

Authors: J. Zhou, D. Dou, P. Pinson (2022)

Venue: Scientific Data | **DOI:** 10.1038/s41597-024-03487-7

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

134-turbine SCADA archive with spatial coordinates, pitch angles, and abnormal status flags

Empirical Scale & Experimental Dataset

Longyuan Power 134-turbine wind farm, 245 days of 10-min records

Relevance to IEEE TSTE & Sustainable Energy Systems

Primary empirical benchmark dataset evaluated in this study

Identified Gaps & Operational Vulnerabilities

KDD Cup competition formulation used symmetric RMSE/MAE, masking asymmetric reserve risk

Role in Manuscript Defense & Reviewer #2 Proof

Source of WTB empirical evidence; proves evaluation strictly adheres to official mask rules